

Evaluation of a polyclonal antibody immunoassay for detection and quantification of *Macrophomina phaseolina*

A. K. SRIVASTAVA and D. K. ARORA*

Laboratory of Applied Mycology, Centre of Advanced Study in Botany, Banaras Hindu University, Varanasi 221 005, India

Detection and quantification of *Macrophomina phaseolina*, causal agent of charcoal/dry root rot disease in many crop plants, was carried out using the ELISA serological technique. Polyclonal antisera raised to water soluble extracts of mycelium, the residual water insoluble mycelial materials or ribosomal proteins were evaluated for specificity and cross-reactivity with 16 common soil fungi by ODD and DAS–ELISA. Soluble and cell wall antisera exhibited strong cross-reactivity with most of the fungal isolates. Ribosomal antibodies were less reactive to common soil fungi except *Fusarium oxysporum* f.sp. *ciceri*. Mycelial antigens of *M. phaseolina* on chickpea roots were detectable with DAS–ELISA at a minimum concentration of 10 ng g^{-1} at 1 : 100 root : buffer dilution. Quantitative estimation of *M. phaseolina* on roots was evaluated by ELISA under different temperatures and moisture conditions, and in soil amended with a potential antagonist (*Trichoderma harzianum* 25-92). A significant reduction in ELISA values was observed in *T. harzianum*-amended treatments. This method may be useful for detection and rapid screening of *M. phaseolina* under different environmental conditions.

INTRODUCTION

Macrophomina phaseolina is a soil-borne fungal pathogen that causes charcoal rot/dry root rot in more than 400 crop plants in different parts of the world (Mihail *et al.*, 1988). The colonization of seed, soil and root surface, distribution and hosts, longevity in soil and competitive saprophytic ability of this pathogen has been extensively studied (Young *et al.*, 1982; Kaiser & Das 1988; Das & Sanker, 1990; Pande *et al.*, 1990; Mukherjee, 1993). Charcoal/dry root rot of chickpea (*Cicer arietinum*), caused by *M. phaseolina*, is one of the most common diseases in south-east Asian countries and severely limits chickpea production (ICRISAT Annual Report 1986; FAO, 1990). Significant reduction (25–70%) in chickpea production is caused by this disease in India (Ahmad *et al.* 1986; Pandey & Singh, 1990; Ahmad & Mouselli, 1990). There is an urgent need to increase chickpea production in these countries to meet the increasing demand for pulses as prime sources of proteins.

* To whom correspondence should be addressed at Applied Microbiology Laboratory, Department of Botany, Banaras Hindu University, Varanasi 221 005, UP, India.

Accepted 24 April 1997.

The characterization and identification of isolates of *M. phaseolina* is usually based on morphological criteria. The morphological differences in *Macrophomina* are minor with virtually no differences in sclerotial morphology but isolates can be distinguished by differences in pathogenicity. Traditional detection from soil, seeds, diseased roots or the collar region is based on baiting techniques and culture on semiselective media (Cloud & Rupe, 1991). Root colonization can be monitored by estimating colony forming units (CFU) per unit area of root, but this method is labour intensive, difficult to interpret, needs taxonomic expertise and requires at least 5–10 days before results can be analysed. Such methods cannot be used to study the interaction between different isolates of *M. phaseolina*, or with ecological studies where a small quantity of inoculum has to be detected. Biochemical methods such as protein electrophoresis and fatty acid and isozyme profiles have also been applied for the characterization and differentiation of fungal taxa (Buth, 1984; Faris *et al.*, 1986; Micales *et al.*, 1992). However, the use of these techniques is limited to taxonomic purposes and to identification and detection of the fungi from pure culture. It would be advantageous to develop a rapid and accurate technique for identification and quantification of this fungus from soil or diseased plant parts.

Immunological methods, because of their high specificity and sensitivity, may provide a possible solution for the detection and quantification of plant pathogenic fungi (Clark, 1981; Kitagawa *et al.*, 1989; Epavier & Alabouvette 1994; Jamaux & Spire, 1994), and are reliable as quantitative techniques for ecological studies (Harrison *et al.*, 1990; Balesdent *et al.*, 1995). Immunological methods can detect very small quantities of fungal inoculum on roots, even under symptomless conditions (Koppel & Schots 1995) and can also be applied to study interactions between pathogen and antagonist in root or soil (El-Nashaar *et al.*, 1986).

Polyclonal antisera were raised from whole cells, wall components, soluble proteins, and ribosomes of different fungal species (Savage & Sall, 1981; Johnson *et al.*, 1982; Hardham *et al.*, 1986; Gleason *et al.*, 1987; Takenaka, 1992). Though there are reports that polyclonal antisera to fungi generally lack specificity to ELISA (enzyme-linked immunosorbent assay) and cross-react with both related and unrelated fungal species (Dewey *et al.*, 1991), some workers demonstrated that species-specific polyclonal antibodies are present in antisera raised against fungi and can be useful for detection and quantification purposes (Adams & Butler, 1979; Jamaux & Spire, 1994). However, there are no reports of the use of serological techniques to detect and quantify *M. phaseolina* in soil or on root surfaces. The aim of this study was to assess the usefulness of polyclonal antisera, raised against different antigens, for detection and quantification of *M. phaseolina* on chickpea roots under different environmental conditions using antisera raised to water-soluble extracts of mycelium, the residual water-insoluble mycelial material or ribosomal proteins. Cross-reactivity of the antisera with related and non-related fungal isolates was also evaluated.

MATERIALS AND METHODS

Media and growth conditions

Isolates of *M. phaseolina* were obtained from different parts of India, and stored on silica gel at 4°C prior to their use (Table 1). Isolates of *T. harzianum* and *M. phaseolina* were grown and maintained on synthetic media (SM; Okon *et al.*, 1973). Other fungal isolates were grown on acidified potato dextrose agar (APDA, pH 5.8).

Disease development and root colonization

Chickpea (*Cicer arietinum*) cv. Desi (Tarai Development Corporation, India) was used throughout

Table 1 Origin and source of soil microorganisms

Fungal isolates	Source
<i>Aspergillus niger</i>	Chickpea field ^a
<i>A. terreus</i>	Chickpea field
<i>Alternaria alternata</i>	Chickpea field
<i>Cochliobolus sativus</i>	Chickpea field
<i>Epicoccum purpurascens</i>	Chickpea field
<i>Fusarium oxysporum</i> f.sp. <i>ciceri</i>	Chickpea field
<i>F. udum</i>	IARI ^b
<i>Humicola grisea</i>	Chickpea field
<i>Macrophomina phaseolina</i> 20-92	LAMCC ^c
<i>M. phaseolina</i> 21-92	LAMCC
<i>M. phaseolina</i> 22-92	LAMCC
<i>M. phaseolina</i> 23-92	LAMCC
<i>Penicillium citrinum</i>	LAMCC
<i>Rhizoctonia solani</i>	Tomato field ^d
<i>Trichoderma harzianum</i> 25-92	Chickpea field
<i>T. harzianum</i> 29-92	Chickpea field

^a Isolated from chickpea field soil near Varanasi in 1992.

^b Indian Agricultural Research Institute, New Delhi.

^c Laboratory of Applied Mycology Culture Collection.

^d Tomato field near Varanasi.

the study. Autoclaved sterilized sandy loam soil (8.5% silt, 22% clay, 69.5% sand, 0.7% OM, and pH 6.9) was mixed with *M. phaseolina*–maize meal inoculum (5% w/w) and incubated for 1 week at 25–28°C to establish the population in soil. Conidia of a known antagonist, *T. harzianum* 25-92 (Srivastava *et al.*, 1996), collected from 6- to 10-day-old cultures, were washed 2–3 times with 50 mM Na-phosphate buffer (PB; pH 7) by centrifugation (1000 *g* for 10 min) and brought to a final concentration of 10⁹ CFU mL⁻¹. The suspension was mixed in the soil previously infested with *M. phaseolina*, and the inoculated soil was equilibrated for 2 days at 25–28°C. Surface-disinfected chickpea seeds (1% NaOCl for 2 min) were sown in inoculated soil in plastic pots (500 g soil per pot; two seeds per pot; 15 × 15 cm). Control plants were grown in pots containing sterilized soil. The pots were watered (c. 70% WHC) and incubated in the greenhouse under cool fluorescent light (photosynthetic irradiance of 350–400 µE m⁻² s⁻¹; 14-h photoperiod). Plants were uprooted at intervals of 7 days until 28 days, and assayed for charcoal rot (Bull *et al.*, 1991). Symptoms on the entire root was recorded on the scale: 1, healthy plant, no disease; 2, brown lesions on collar region of root; 3, dark brown to charcoal spots on entire root; 4, complete rotting of root. The root segments (c. 1 g) were assayed for

colonization of *M. phaseolina* by double antibody sandwich ELISA (DAS–ELISA).

Production of antigen and antisera

The antigens were prepared from water soluble extracts of mycelium, the residual water insoluble mycelial materials or ribosomal proteins of *M. phaseolina* 20-92 obtained from 6-day-old static SM cultures. The mycelial mat was filtered through muslin, washed twice with PB (pH 7), and lyophilized. Lyophilized mycelium (1 g) was ground with liquid nitrogen. Phosphate buffer saline (PBS; 10 mL) containing 0.1% cysteine, 0.1% ascorbic acid, and 17% sucrose was added to ground mycelium (El-Nashaar *et al.*, 1986) and the suspension was centrifuged at 45000 *g* for 30 min at 4°C. The supernatant was concentrated over polyethylene glycol 6000 (Hi Media Laboratories, India), dialysed against PBS (0.1 M, pH 7) for 24 h. The pellet was resuspended in 1 mL of PBS. The protein content of both supernatant and pellet was determined (Lowry *et al.*, 1951) and adjusted to 1 mg mL⁻¹ in PBS. These suspensions were stored at -20°C and used as soluble protein and cell wall antigens, respectively.

Ribosomes were obtained from mycelial homogenate of *M. phaseolina* grown for 6 days in SM (Takenaka & Yoshino, 1987). The homogenate was ultracentrifuged through a sucrose pad and treated with bovine pancreatic ribonuclease-A (Sigma, St Louis, MO, 1 mg mL⁻¹) at 4°C for 30 min. The suspension was layered on a sucrose density gradient (10–40% w/v) in 20 mM Tris-HCl, 20 mM KCl and 10 mM MgCl₂ (pH 7.5), and centrifuged at 40000 *g* for 12 h at 4°C. The ribosome pellet was collected, resuspended in gradient buffer and stored at -20°C until use.

Rabbits (white; 2) were immunized by subcutaneous injection of 1 mL of either water-soluble extracts of mycelium, the residual water-insoluble mycelial materials or ribosomal proteins antigen (1 mg mL⁻¹ protein, or 20 units of ribosomes A₂₆₀) emulsified with an equal volume of Freund's incomplete adjuvant (Sigma). Four injections were given at intervals of 8 days followed by a booster injection 4 weeks later. The blood was collected from the marginal vein of the ear 1 week after the booster injection. Antisera were separated from coagulated blood by centrifugation at 3000 *g* for 5 min and stored at -20°C until use.

Ouchterlony double diffusion test

Ouchterlony double diffusion (ODD) tests were

performed in plastic dishes (90 mm) containing 12 mL of 0.75% bacto-agar no. 1 (Difco Laboratories, Detroit, MI, USA). A central well (5 mm diameter) surrounded by 6–8 peripheral wells 5 mm away, were cut, using a sterilized stainless steel cork borer. Antisera raised from water-soluble extracts of mycelium, the residual water-insoluble mycelial materials or ribosomal proteins were placed separately (15 µL) in central wells. The outer wells contained 10 µL of the soluble mycelial antigens obtained from 15 fungal isolates (Table 1). Preparation of these antigens was similar to that described for *M. phaseolina*. The plates were sealed with parafilm and incubated at 3–4°C for 1–3 days. Precipitation lines were observed and recorded as: 1, complete fusion; 2, partial fusion (cross-reaction); and 3, no fusion (no reaction).

Double antibody sandwich ELISA

The IgG fraction (γ globulin) from ribosomal antiserum was purified on a protein A sepharose (Pharmacia, Uppsala, Sweden) column (Harlow & Lane, 1988). IgG (1 mg) in 1 mL sucrose buffer was mixed with 1.5 mg alkaline phosphatase (Sigma) to prepare the IgG alkaline phosphatase conjugate. The solution was dialysed against 0.06% glutaraldehyde (E. Merck, India) in PBS at 25°C for 6 h and then further dialysed at 4°C for 24 h, and stored in 50% (w/v) glycerol at -20°C.

The DAS–ELISA was conducted in microtiter plates. These were washed three times with PBS-T (0.05% Tween 20 in PBS) between each step, and for each step the reagent volume was 100 µL per well. The plates were coated with 5 mg mL⁻¹ IgG in 0.05 M carbonate buffer (pH 9.6) and incubated for 1.5 h at 37°C. The antigens (15 fungal isolates) diluted in PBS-T-PVP (2% polyvinyl pyrrolidone in PBS-T) were then added to the plates and incubated for 24 h at 25°C with constant shaking. The conjugant diluted in PBS-T-PVP (1 : 1000 v/v) was added and maintained overnight at 4°C. The plates were washed and finally the substrate (1 mg mL⁻¹ *p*-nitrophenyl phosphate in diethanolamine–carbonate buffer, pH 9.8) was added and optical densities (OD) were read in an ELISA reader at 405 nm 5–6 times at 25°C. The ribosomal antiserum (1 mL) was incubated overnight with the mycelial extract (1 mL; 1 mg protein) of *Fusarium oxysporum* f. sp. *ciceri* for cross absorption and antiserum obtained after absorption was again tested by DAS–ELISA.

Quantitative DAS–ELISA

For quantification of colonization of *M. phaseolina*

on chickpea roots, frozen root segments were ground in a tissue homogenizer in PBS-T-PVP at a ratio of 1 : 25 (w/v). The samples were centrifuged (15000 *g* for 10 min; 4°C) and the supernatant was diluted twice in PBS-T-PVP to obtain at least one value within the standard protein-ELISA curve. The standard curve was prepared previously by plotting the ELISA values of the diluted protein extract of *M. phaseolina* (antigen; 10–0.005 mg mL⁻¹ in PBS-T-PVP). The ELISA values were converted into fungal protein concentration using the regression curve obtained with standard protein.

Effects of factors such as incubation time and

temperature, which greatly influence the survival of *M. phaseolina* in soil and colonization of roots, were also determined by DAS–ELISA. To evaluate the effect of inoculum concentration of *T. harzianum*, the chickpea plants were grown in pots containing *M. phaseolina*-infested soil inoculated with *T. harzianum* 25-92 at 10³–10⁹ CFU. Proteins from infected roots were prepared and ELISA values were recorded. Effects of soil moisture and temperature on root colonization were studied by incubating the pots at 0, –5, –10 kPa moisture at 28°C, and at 20, 25, 30, 35 and 40°C at –5 kPa. Moisture levels were maintained by adding an

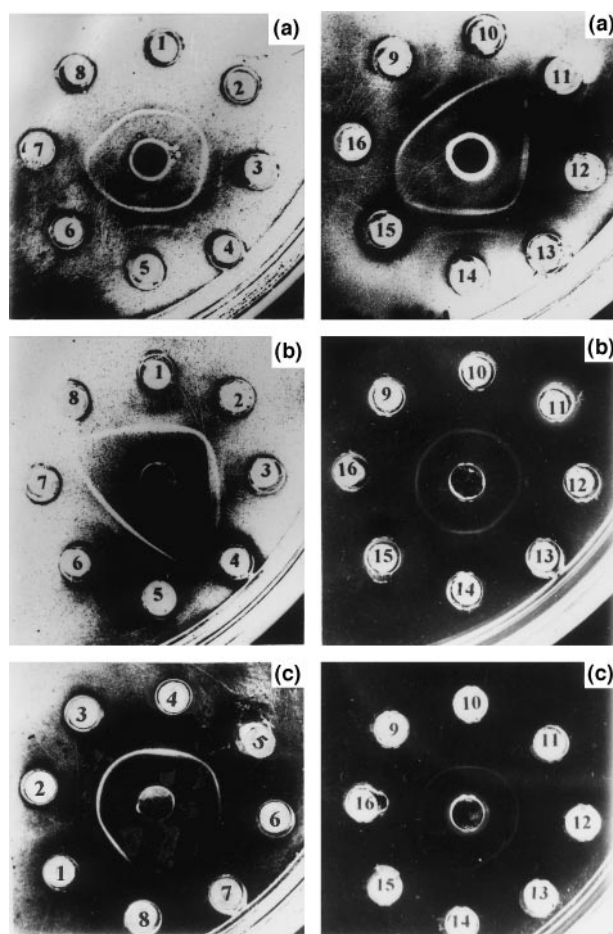


Fig. 1 Ouchterlony double diffusion reactions of antisera raised from water-soluble extracts of mycelium (a), the residual water-insoluble mycelial materials (b) or ribosomal proteins (c) to antigens of different fungal isolates. Central well contains 15 mL of antisera and outer wells contain antigen of *Macrophomina phaseolina* 20-92, 21-92, 22-92, 23-92 (1–4), *Fusarium oxysporum* f.sp. *ciceri* (5), *F. udum* (6), *Humicola grisea* (7), *Alternaria alternata* (8), *Aspergillus niger* (9), *A. terreus* (10), *Trichoderma harzianum* 25-92, 29-92 (11, 12), *Penicillium citrinum* (13), *Cochliobolus sativus* (14), *Epicoccum purpurascens* (15) and *Rhizoctonia solani* (16).

amount of water calculated from a moisture retention curve (Filonow & Arora, 1987) every 2 days. After 28 days of incubation, plants were uprooted and fungal antigens from infected roots were assayed for ELISA values.

The experiment was repeated twice for reproducibility; data from one set of experiments are presented. The differences between mean values in each set were used to calculate standard deviations.

RESULTS

Antisera raised from soluble mycelial proteins, the residual water-insoluble mycelial material or ribosomes of *M. phaseolina* 20-92 were tested against antigens of 16 related and unrelated fungal isolates (Fig. 1). The antiserum from soluble mycelial protein cross-reacted with *Aspergillus niger*, *A. terreus*, *Fusarium oxysporum* f.sp. *ciceri*, *F. udum*, *Penicillium citrinum*, *Cochliobolus sativus*, *Humicola grisea*, and *Epicoccum purpurascens*, and all four isolates of *M. phaseolina*. A sharp precipitation band was observed midway between antigens and antisera when tested against these fungal isolates

(Fig. 1a). A weak diffused band was also recorded with *Rhizoctonia solani* and *Alternaria alternata* (Fig. 1a). Similar results were also obtained with the antisera raised from the residual water-insoluble mycelial materials except that a very weak diffused band was observed with *A. niger*, *A. terreus*, *P. citrinum*, *C. sativus*, *H. grisea*, *E. purpurascens*, *R. solani* and *A. alternata* (Fig. 1b). Isolates of *F. oxysporum* f.sp. *ciceri*, *F. udum* and *M. phaseolina* exhibited a sharp band. In contrast, antisera raised against ribosomal protein reacted positively only with the *M. phaseolina* isolates, *F. udum* and *F. oxysporum* f.sp. *ciceri* (Fig. 1c). Other fungal isolates did not show cross-reactivity.

Specificity of DAS-ELISA

Cross-reactivity of ribosomal antisera was further evaluated by DAS-ELISA. An intense cross-reaction was observed between four isolates of *M. phaseolina*, *A. niger*, *F. oxysporum* f.sp. *ciceri*, and to a lesser extent with *F. udum* (Fig. 2). Very weak or no response was noticed with the other isolates including *T. harzianum* 25-92 and 29-92.

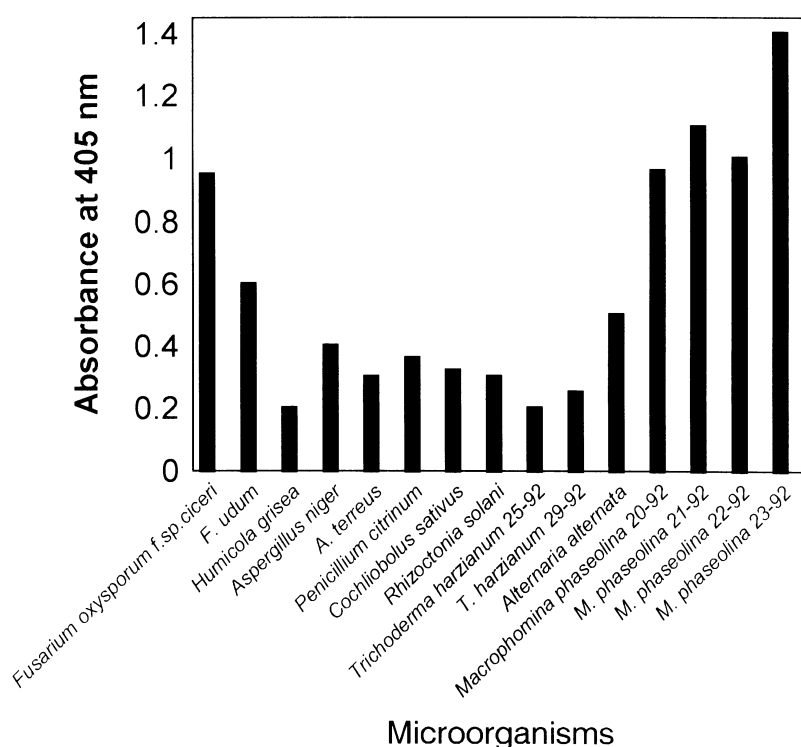


Fig. 2 ELISA response to the protein extracts of 16 fungal isolates. The protein extracts were adjusted to $1 \mu\text{g mL}^{-1}$.

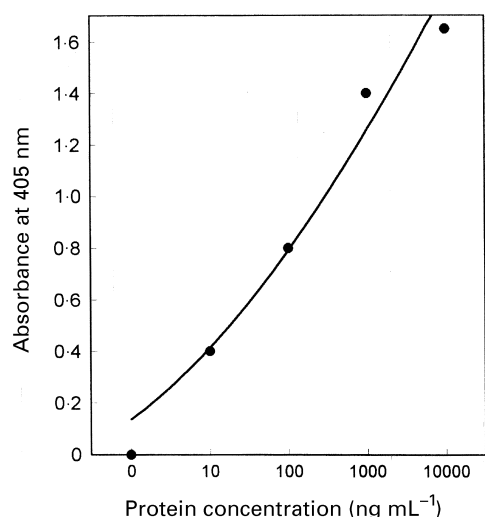


Fig. 3 Dose-response curve (regression) to *Macrophomina phaseolina* using DAS-ELISA.

The cross-absorption of antisera with *F. oxysporum* f.sp. *ciceri* also did not react with the *M. phaseolina* isolates (data not shown). Therefore, cross-reactivity could not be reduced or eliminated through absorption and further experiments were conducted using ribosomal antisera.

Detection of colonization

A regression curve was plotted between *M. phaseolina* proteins (antigens) and absorbance at 405 nm (Fig. 3). An absorbance of 1.4 was recorded with fungal protein of 1000 ng mL⁻¹, whereas it was 0.4

Table 2 Estimation of *Macrophomina phaseolina* protein on diseased roots by ELISA

Days ^a	<i>Trichoderma harzianum</i> 25-92	A _{405 nm}	Concentration of mycelium (ng g ⁻¹ root)	Disease index
7	+	0.3	10	1
	-	0.5	45	1
14	+	0.4	22	1
	-	1.0	500	3
21	+	0.5	45	1
	-	1.25	980	4
28	+	0.45	40	1
	-	1.2	800	4

^aDays after sowing chickpea. +, soil amended with *T. harzianum*; -, soil not amended with *T. harzianum*.

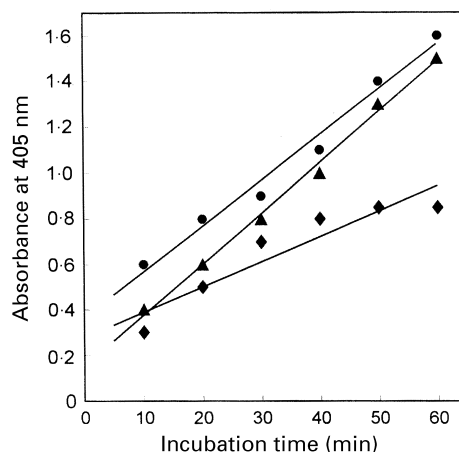


Fig. 4 Development of colour in ELISA with root extract containing water-soluble mycelial proteins at different dilutions. ♦, undiluted root extract; ▲, 1:10 root extract:buffer; ●, 1:100 root extract:buffer (w/v). Ribosomal proteins were used to raise the antisera.

at 10 ng mL⁻¹ protein. The ELISA values obtained between 0.4 and 1.4 can therefore be converted to the protein content of the sample (Fig. 3). Colonization of chickpea roots by *M. phaseolina* 20-92 was quantified by the ELISA values using a regression curve (Table 2; Fig. 4). Optimum dilution for quantification of *M. phaseolina* on the root was 1:100, whereas 1:10 and undiluted root extract

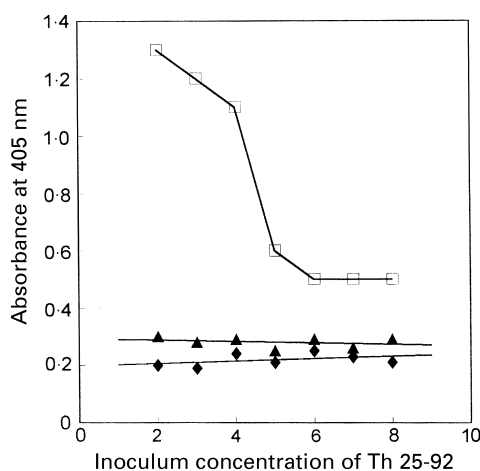


Fig. 5 ELISA values of extracts of roots colonized by *Macrophomina phaseolina* at different inoculum concentration of *Trichoderma harzianum* 25-92 in soil. ♦, buffer; ▲, healthy root extracts (control); □, infected root extract. Ribosomal proteins were used to raise the antisera.

strongly reduced the ELISA values (Fig. 4). ELISA values were significantly higher for roots colonized with *M. phaseolina* as compared to the healthy uninoculated control (Table 2). However, maximum colonization (protein content 980 ng g^{-1} root) was recorded from roots of plants 21 days old, equivalent to the colonizing population of about $3.9 \log \text{ CFU cm}^{-1}$ root (data not shown). No further increase in ELISA values was obtained 21 days after colonization. A marked reduction in colonization by *M. phaseolina* was recorded in soil amended with *T. harzianum* 25-92 (Table 2). Though charcoal rot symptoms were not observed in *T. harzianum* 25-92-amended treatments, a small quantity of *M. phaseolina* ($10\text{--}50 \text{ ng g}^{-1}$ root) was detected from the roots.

Factors affecting colonization by *M. phaseolina*

Studies were performed to detect the root colonization by *M. phaseolina* in the presence of *T. harzianum*

25-92, a potential biocontrol agent of charcoal rot of chickpea (Srivastava *et al.*, 1996). A significant ($P=0.05$) decrease in ELISA values was recorded with increasing inoculum concentration of *T. harzianum* 25-92. Maximum ELISA value (1.3) was recorded at an inoculum concentration of $2 \log \text{ CFU}$, decreasing at a steady rate up to $5 \log \text{ CFU}$ (1.05), and thereafter a sharp decrease in ELISA values was obtained up to $7 \log \text{ CFU}$ (0.5) (Fig. 5). No significant ($P=0.05$) change in ELISA values was recorded with further increases in concentration of *T. harzianum*.

Soil moisture and temperature also affected the ability of *M. phaseolina* to colonize chickpea roots (Fig. 6a,b). In the treatments not amended with *T. harzianum* 25-92, sharp increases in ELISA values were recorded when moisture levels decreased from -1 to -10 kPa . However, a similar trend of absorbance was also recorded in antagonist-amended treatments, absorbance (405 nm) values were about two-fold less than those of the control (Fig. 6a).

Infection of chickpea roots by *M. phaseolina* was evaluated at different temperatures ($20, 25, 30, 35$ and 40°C). Maximum absorbance ($1.25\text{--}1.35$) was recorded at $30\text{--}35^\circ\text{C}$ in unamended treatments. At lower temperatures ($15\text{--}20^\circ\text{C}$), ELISA values were much less ($0.4\text{--}0.6$). ELISA values decreased by 2- to 2.3-fold in the treatments amended with *T. harzianum* (Fig. 6b).

DISCUSSION

This study was aimed to develop a rapid and accurate ELISA-serological technique for the detection and quantification of *M. phaseolina* on chickpea roots under greenhouse conditions. Previous studies on colonization of roots of host plants by *M. phaseolina* and on host-pathogen interactions were based on indirect isolation techniques, which influence the number of colony forming units (CFU) per unit of roots. Gerik *et al.* (1987) developed a serological staining method for direct observation and quantification of root colonization by *Verticillium* and *Fusarium* spp. Rapid and accurate techniques based on quantitative ELISA were successfully developed to estimate the colonization of roots by several soil-borne fungal pathogens (El-Nashaar *et al.*, 1986; Kumar *et al.*, 1993; Balesdent *et al.*, 1995). However, there are no reports describing the use of serological techniques to detect and quantify *M. phaseolina* in soil or on root surfaces.

Antibodies against fungal pathogens have been raised from soluble proteins, crude cell components, fungal homogenate, culture fluids, and ribosomal proteins (Savage & Sall, 1981; Hardham *et al.*, 1986;

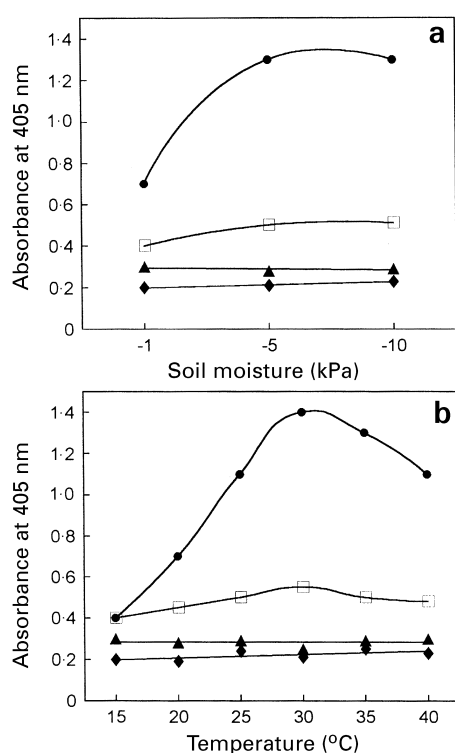


Fig. 6 ELISA values for the colonized roots at different moisture levels (a) and temperatures (b). $\blacklozenge$, buffer; $\blacktriangle$, healthy root extracts (control); $\square$, with *Trichoderma harzianum* 25-92; $\bullet$, without *T. harzianum* 25-92. Ribosomal proteins were used to raise the antisera.

Gleason *et al.*, 1987; Reddick & Collins, 1988; Lange *et al.*, 1989). In this study, antibodies were raised to soluble mycelial proteins, insoluble residual materials or ribosomal proteins of *M. phaseolina*. Antisera raised from soluble mycelial proteins has limited significance for detection of *M. phaseolina* *in vitro* and on root surfaces (Fig. 1). The cell wall proteins also showed cross-reactivity to many common soil fungi. Antibodies developed against ribosomal proteins had greater specificity as compared with antibodies raised against soluble and cell wall proteins. Ribosomal antisera showed cross-reaction with four different isolates of *M. phaseolina* and *Fusarium oxysporum* f.sp. *ciceri* (Fig. 1). Similarly, Takenaka (1992) also found that polyclonal antisera raised against ribosomes were genus-specific. This could be because they are present in all cellular organisms and contain highly conserved and moderately antigenically simpler proteins. Other workers (Musgrave *et al.*, 1984; Mohan, 1988; Harrison *et al.*, 1990; Dewey *et al.*, 1991; Velicheti, 1993) also demonstrated that polyclonal antisera raised against one fungal species are often genus specific and specificity can rarely be improved through the cross-absorption technique (Ricker *et al.*, 1991).

The specificity of cell wall and ribosomal antibodies was further evaluated by DAS-ELISA. Cell wall antibodies were not found specific at genus level, whereas ribosomal antibodies showed a weak specificity to *M. phaseolina* (Fig. 2). All isolates other than *M. phaseolina* that displayed strong cross-reactivity in the ODD test also showed a positive ELISA response. In this study, the ribosomal antibodies from *M. phaseolina* also cross-reacted with *F. udum* and *F. oxysporum* f.sp. *ciceri* indicating serological identity and presence of common epitopes. Thus, these results indicate that antibodies raised from ribosomal proteins are specific at genus level and could be used to detect *M. phaseolina* *in vitro* and in sterilized soil. The results are in agreement with previous studies (Goff & Begueret, 1984; Partridge *et al.*, 1984; Takenaka, 1992) suggesting that ribosomal proteins may show significant immunological differences among different genera because of their highly conserved nature. These proteins do not differ qualitatively at different stages of development. The results presented here indicate that chickpea roots infected by *M. phaseolina* can be detected by DAS-ELISA. This technique seems to be sensitive enough to measure quantitatively the colonization at a concentration of 10 ng mL^{-1} proteins in root homogenates. ELISA values in

control assays (buffer and healthy root homogenate) were very low. Therefore, quantification of a population of *M. phaseolina* on roots can be achieved using a protein absorbance curve (Fig. 3). In these experiments, extreme sensitivity of ELISA was evident as colonization of *M. phaseolina* could be detected as early as 1 week after planting (Fig. 4).

These results also demonstrate that ribosomal-specific antibodies may be useful for the detection of *M. phaseolina* under different environmental conditions in greenhouse assays. In the experiment performed at different temperatures and moisture levels, maximum root colonization occurred at 30°C and -5 kPa soil moisture. However, the ELISA values decreased significantly ($P=0.05$) in the treatments amended with *T. harzianum* (Fig. 5). Different inoculum concentrations of the antagonist (10^3 – 10^8 CFU g^{-1} soil) yielded different ELISA values for *M. phaseolina*. The maximum significant ($P=0.05$) decrease in ELISA values was recorded at an inoculum concentration of 7 log CFU indicating that the population of *M. phaseolina* was minimal at this concentration of the antagonist. The results also demonstrate the competence of *T. harzianum* in preventing colonization of *M. phaseolina* at different moisture levels and temperatures (Fig. 6). Overall, this study reveals that ribosomal antisera can detect *M. phaseolina* in an *in vitro* system and in the greenhouse. Very small populations of *M. phaseolina* on the root can be detected by ELISA and relative activity of *T. harzianum* can be evaluated from differences in the amount of antigen present in root homogenates of antagonist-amended and control treatments. Therefore, ELISA could be a useful tool for evaluation and detection of *M. phaseolina* and for studies of antagonist–pathogen interaction under laboratory conditions. The technique may also be useful for the detection of disease-resistant plants in breeding programmes. Although the polyclonal antisera raised from *M. phaseolina* ribosomes worked well in this system, it could not differentiate between *M. phaseolina* isolates and different species of *Fusarium*. Therefore, further studies are needed to increase the specificity of antisera by developing specific monoclonal antibodies that may be helpful in the detection and quantification of the pathogen in the field.

ACKNOWLEDGEMENT

We thank the University Grants Commission, New Delhi for a research fellowship to AKS.

REFERENCES

- Adams Jr GC, Butler EE, 1979. Serological relationships among anastomosis groups of *Rhizoctonia solani*. *Phytopathology* **69**, 629–33.
- Ahmad Q, Abu MAQ, Mohammad A, 1986. Losses in yield due to *Rhizoctonia* root rot of chickpea in Bihar. *Indian Phytopathology* **39**, 590–2.
- Ahmad EM, Mouselli MN, 1990. Wilt and root rot of chickpea in southern Syria. *Arab Journal of Plant Protection* **8**, 60–7.
- Balesdent MH, Destheux C, Gall PR, Rouxel T, 1995. Quantification of *Leptosphaeria maculans* growth in cotyledons of *Brassica napus* using ELISA. *Journal of Phytopathology* **143**, 65–73.
- Bull CT, Weller DM, Thomashow LS, 1991. Relationship between root colonization and suppression of *Gaeumannomyces graminis* var. *tritici* by *Pseudomonas fluorescens* strain 2-79. *Phytopathology* **81**, 945–59.
- Buth DG, 1984. The application of electrophoretic data in systematic studies. *Annual Review of Ecology and Systematics* **15**, 501–22.
- Clark MF, 1981. Immunosorbent assay in plant pathology. *Annual Review of Phytopathology* **19**, 83–106.
- Cloud GL, Rupe JC, 1991. Comparison of three media for enumeration of sclerotia of *Macrophomina phaseolina*. *Plant Diseases* **75**, 771–2.
- Das ND, Sankar GRM, 1990. Effect of root rot disease caused by *Macrophomina phaseolina* (Tassi) Goid. of castor (*Ricinus communis*) by statistical modelling of cultural practices of carbendazim applications. *Indian Journal of Mycology and Plant Pathology* **20**, 234–40.
- Dewey M, Evans D, Colemann J, Priestley R, Hull R, Horsley D, Hawes C, 1991. Antibodies in plant science. *Acta Botanica Neerlandica* **40**, 1–27.
- El-Nashaar HM, Moore LW, George RA, 1986. Enzyme linked immunosorbent assay quantification of initial infection of wheat by *Gaeumannomyces graminis* var. *tritici* as moderated by biocontrol agents. *Phytopathology* **76**, 1319–22.
- Eparvier A, Alabouvette C, 1994. Use of ELISA and GUS-transformed strains to study competition between pathogenic and nonpathogenic *Fusarium oxysporum* for root colonization. *Biocontrol Science and Technology* **4**, 35–47.
- FAO, 1990. *Production Year Book 1989*. Rome, Italy: Food and Agriculture Organization of the United Nations.
- Faris MA, Sabo FE, Cloutier Y, 1986. Intraspecific variation in gel electrophoresis patterns of soluble mycelial proteins of *Phytophthora megasperma* isolated from alfalfa. *Canadian Journal of Botany* **64**, 262–5.
- Filonow AB, Arora DK, 1987. Influence of soil matrix potential on ¹⁴C-exudation from fungal propagules. *Canadian Journal of Botany* **65**, 2084–9.
- Gerik JS, Lommel SA, Huisman OC, 1987. A specific serological staining procedure for *Verticillium dahliae* in cotton root tissue. *Phytopathology* **77**, 261–5.
- Gleason ML, Gabriel SA, Ferriss RS, 1987. Serological detection of *Phomopsis longicolla* in soybean seeds. *Phytopathology* **77**, 371–5.
- Goff V, Le Begueret J, 1984. Immunological comparison of individual ribosomal proteins in six species of the genus *Podospora*. *Molecular Genetics* **193**, 143–8.
- Hardham AR, Suzaki E, Perkin JL, 1986. Monoclonal antibodies to isolate species and genus specific components on the surface of zoospores and cysts of fungus *Phytophthora cinnamomi*. *Canadian Journal of Botany* **64**, 311–25.
- Harlow E, Lane D, 1988. *Antibodies: A Laboratory Manual*. New York, NY, USA: Cold Spring Harbor Laboratory Press.
- Harrison JG, Barker H, Lowe R, Rees EA, 1990. Estimation of amounts of *Phytophthora infestans* mycelium in leaf tissue by enzyme linked immunosorbent assay. *Plant Pathology* **39**, 274–7.
- ICRISAT Annual Report (1986). *Sorghum Mycorrhizas and Diseases*. Pattencheru, India: International Crops Research Institute for Semi-Arid Tropics. 35–41.
- Jamaux I, Spire D, 1994. Development of polyclonal antibody based immunoassay for the early detection of *Sclerotinia sclerotiorum* in rapeseed petals. *Plant Pathology* **43**, 847–62.
- Johnson MC, Pirone TP, Siegel MR, Varney DR, 1982. Detection of *Epichloe typhina* in tall fescue by means of enzyme linked immunosorbent assay. *Phytopathology* **72**, 647–50.
- Kaiser SAKM, Das SN, 1988. Physical factors that influence the growth and spread of charcoal rot pathogen (*Macrophomina phaseolina*) infecting maize. *Journal of Phytopathology* **123**, 47–51.
- Kitagawa T, Sakamoto Y, Furumi K, Ogura H, 1989. Novel enzyme immunoassay for specific detection of *Fusarium oxysporum* f. sp. *cucumerinum* and for general detection of various *Fusarium* species. *Phytopathology* **79**, 162–5.
- Koppel MMV, Schots A, 1995. Monoclonal antibody based double antibody sandwich ELISA for detection of *Verticillium* spp. in ornamentals. *Phytopathology* **85**, 608–12.
- Kumar VU, Meera MS, Shishupala S, Shetty HS, 1993. Quantification of *Sclerospora graminicola* in tissues of *Pennisetum glaucum* using an enzyme linked immunosorbent assay. *Plant Pathology* **42**, 251–5.
- Lange L, Heide M, Hobolth L, Olson LW, 1989. Serological detection of *Plasmodiophora brassicae* by dot immunobinding and visualization of serological reaction by scanning electron microscopy. *Phytopathology* **79**, 1066–71.
- Lowry OH, Rosenberg NH, Farr AL, Randall RH, 1951. Protein measurement with the folin phenol reagent. *Journal of Biological Chemistry* **193**, 265–75.
- Micales JA, Bonde MR, Patterson GL, 1992. Isozyme analysis in fungal taxonomy and molecular genetics. In: Arora DK, Elander RP, Mukerji KG, eds. *Handbook of Applied Mycology* vol. 4. New York, NY, USA: Marcel Dekker, Inc., 57–79.
- Mihail JD, Young DJ, Alcorn SM, 1988. *Macrophomina phaseolina*: a plant pathogen of concern in arid lands. In: Whitehead EE, Hutchin CF, Timmerman BN, Varady RG, eds. *Arid-Lands: Today and Tomorrow*. New York, NY, USA: West View Press, 1305–10.

- Mohan SB, 1988. Evaluation of antisera against *Phytophthora fragariae* for detecting the red core disease of strawberries by enzyme linked immunosorbent assay (ELISA). *Plant Pathology* **37**, 206–16.
- Mukherjee PK, 1993. Biological control of collar rot, dry root rot wilt and gray mold of chickpea. *International Crops Research Institute for Semi-Arid Tropics Bulletin* India: A.P. Patancheru.
- Musgrave DR, 1984. Detection of an endophytic fungus of *Lolium perenne* using enzyme linked immunosorbent assay (ELISA). *New Zealand Journal of Agricultural Research* **27**, 283–8.
- Okon Y, Chet I, Henis Y, 1973. Effect of lactose ethanol and cyclohexamide on translocation pattern of radio-active compounds on sclerotia formation in *Sclerotium rolfsii*. *Journal of General Microbiology* **74**, 251–8.
- Pande S, Mughogho LK, Karunakar RI, 1990. Effect of moisture stress, plant population density and pathogen inoculation on charcoal stalk rot of sorghum. *Annals of Applied Biology* **116**, 221–32.
- Pandey G, Singh RB, 1990. Survey of root diseases of chickpea in Allahabad region. *Current Nematology* **1**, 77–8.
- Partridge JE, Nelson PE, Toussoun TA, 1984. Ribosomal proteins of the genus *Fusarium*. *Mycologia* **76**, 533–44.
- Reddick BB, Collins MH, 1988. An improved method for detection of *Acremonium coenophialum* in tall fescue plants. *Phytopathology* **78**, 418–20.
- Ricker RW, Marios JJ, Dlott JW, Bostock RM, Morrison JC, 1991. Immunodetection and quantification of *Botrytis cinerea* on harvested wine grapes. *Phytopathology* **71**, 404–11.
- Savage SD, Sall MA, 1981. Radioimmunosorbent assay for *Botrytis cinerea*. *Phytopathology* **71**, 411–15.
- Srivastava AK, Arora DK, Gupta S, Pandey RR, Lee MW, 1996. Diversity of potential microbial parasites colonizing sclerotia of *Macrophomina phaseolina* in soil. *Biology and Fertility of Soils* **22**, 136–40.
- Takenaka S, 1992. Use of immunological methods with antiribosome serum to detect snow mold fungi in wheat plants. *Phytopathology* **82**, 896–901.
- Takenaka S, Yoshino R, 1987. Detection of snow mold fungi in wheat plants by means of polyacrylamide gel electrophoresis of ribosomal proteins. *Annals of the Phytopathological Society of Japan* **53**, 591–7.
- Velicheti RK, 1993. Immunodetection of *Phomopsis* species in asymptomatic soybean plants. *Plant Disease* **77**, 404–11.
- Young DL, Gilbertson RL, Alcorn SM, 1982. A new record for longevity of *Macrophomina phaseolina* sclerotia. *Mycologia* **74**, 504–5.